Dual Armed Forces of Greater India

Complex Military Field Theory using $\Phi_2(x)$ and $\Phi_{113}(x)$

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Abstract

This treatise proposes a mathematical and institutional framework for studying two complementary defence establishments under the conceptual title "Dual Armed Forces of Greater India." Its formal foundation is the pair of flat cyclotomic polynomials $\Phi_2(x)$ and $\Phi_{113}(x)$, whose algebraic properties are exactly established. Their product illustrates an important distinction: flatness of individual polynomials does not imply flatness of their product.

The proposed Complex Military Field Theory connects algebraic structures with defence economics, public finance, institutional design, strategic stability, welfare economics, risk analysis, cybersecurity, and international relations. The cyclotomic representation is a modelling choice, not an established physical law or a demonstrated theory of military effectiveness. Empirical validation, legal compliance, civilian protection, and transparent resource allocation remain necessary conditions for any practical application.

The treatise ends with "The End"

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1 Introduction

Defence institutions operate within complex environments involving uncertainty, constrained public resources, technological change, geography, diplomacy, and competing national priorities.

The concept of dual armed forces is examined here as a proposed institutional architecture in which two complementary defence organisations have distinguishable responsibilities but share appropriate standards of accountability, civilian oversight, and strategic coordination.

The mathematical foundation consists of two cyclotomic polynomials of prime index. Their algebraic properties provide a simple example of how individual components can possess a common structural property while their combination exhibits a different property.

The central research question is:

$$\textit{How can complementary institutions be coordinated without assuming that structural symmetry guarantees optimal outcomes?} \tag{1}$$

The answer requires contributions from mathematics, economics, statistics, law, political science, engineering, and the social sciences.

2 Mathematical Foundations

2.1 Definition of a cyclotomic polynomial

Let n be a positive integer and let ζ_n denote a primitive nth root of unity. The nth cyclotomic polynomial is defined by

$$\Phi_n(x) = \prod_{\substack{1 \le k \le n \\ \gcd(k,n)=1}} \left(x - e^{2\pi i k/n} \right). \tag{2}$$

Cyclotomic polynomials have integer coefficients and are irreducible over the rational numbers. Their degree is given by Euler's totient function (Apostol, 1976; Washington, 1997).

$$\deg \Phi_n(x) = \varphi(n). \tag{3}$$

For a prime number p, the polynomial simplifies to

$$\Phi_p(x) = \frac{x^p - 1}{x - 1} = \sum_{k=0}^{p-1} x^k. \tag{4}$$

2.2 The polynomial $\Phi_2(x)$

Because 2 is prime,

$$\Phi_2(x) = \frac{x^2 - 1}{x - 1} = x + 1. \tag{5}$$

Its degree is one, and its two coefficients are both equal to one.

2.3 The polynomial $\Phi_{113}(x)$

The integer 113 is prime. Consequently,

$$\Phi_{113}(x) = \frac{x^{113} - 1}{x - 1}. \tag{6}$$

Its expanded form is

$$\Phi_{113}(x) = x^{112} + x^{111} + \cdots + x^2 + x + 1. \tag{7}$$

The degree is 112, and all 113 coefficients are equal to one.

2.4 Flatness

Consider a nonzero polynomial

$$P(x) = \sum_{k=0}^{d} a_k x^k. \tag{8}$$

Its height is defined as

$$H(P) = \max_{0 \le k \le d} |a_k|. \tag{9}$$

A polynomial with integer coefficients is called flat here when every coefficient belongs to $\{-1, 0, 1\}$ and at least one coefficient is nonzero. Equivalently, it has height one.

For the two polynomials under consideration,

$$H(\Phi_2) = 1, \tag{10}$$
$$H(\Phi_{113}) = 1. \tag{11}$$

Both are therefore flat.

Table 1: Algebraic properties of the two cyclotomic polynomials

Property	$\Phi_2(x)$	$\Phi_{113}(x)$
Index	2	113
Degree	1	112
Nonzero terms	2	113
Coefficient set	$\{1\}$	$\{1\}$
Height	1	1
Flatness	Yes	Yes
Irreducible over $\mathbb{Q}$	Yes	Yes

2.5 Multiplication and the loss of flatness

The product is

$$P(x) = \Phi_2(x)\Phi_{113}(x) \tag{12}$$
$$= (x+1)(1 + x + x^2 + \cdots + x^{112}) \tag{13}$$
$$= 1 + 2x + 2x^2 + \cdots + 2x^{112} + x^{113}. \tag{14}$$

Thus,

$$H(P) = 2. \tag{15}$$

The product is not flat. Its coefficients are one at the two endpoints and two at all intermediate powers.

This gives a fundamental mathematical observation:

$$H(P) = H(Q) = 1 \quad \not\Rightarrow \quad H(PQ) = 1. \tag{16}$$

The example is an algebraic fact. Any interpretation of it as a model of institutional complementarity requires separate assumptions and evidence.

3 From Algebra to Complex Military Field Theory

3.1 A proposed institutional interpretation

Let the two institutions be denoted by $\mathcal{D}_1$ and $\mathcal{D}_2$. Their roles must be defined by constitutional law, legislation, civilian authority, and the actual administrative arrangements of the jurisdiction concerned.

For analytical purposes, let their resource and capability states be represented by

$$\mathbf{z}_1(t) \in \mathbb{R}^{m_1}, \qquad \mathbf{z}_2(t) \in \mathbb{R}^{m_2}. \tag{17}$$

The combined state is

$$\mathbf{z}(t) = \begin{pmatrix} \mathbf{z}_1(t) \\ \mathbf{z}_2(t) \end{pmatrix}. \tag{18}$$

A complex-valued analytical variable may be introduced as

$$z(t) = a(t) + ib(t), \qquad a(t), b(t) \in \mathbb{R}. \tag{19}$$

Here, the real and imaginary components can represent two distinct analytical dimensions, such as realised resource use and a modelled uncertainty component. This interpretation is a modelling convention; complex numbers do not by themselves represent physical military capabilities.

A candidate cyclotomic field representation is

$$F(z) = \alpha(t)\Phi_2(z) + \beta(t)\Phi_{113}(z), \tag{20}$$

where $\alpha(t)$ and $\beta(t)$ are explicitly defined, dimensionally compatible coefficients.

This expression is a proposed representation rather than a validated defence model. Its usefulness depends on whether the variables, coefficients, and resulting predictions have operationally meaningful definitions.

3.2 A conceptual architecture

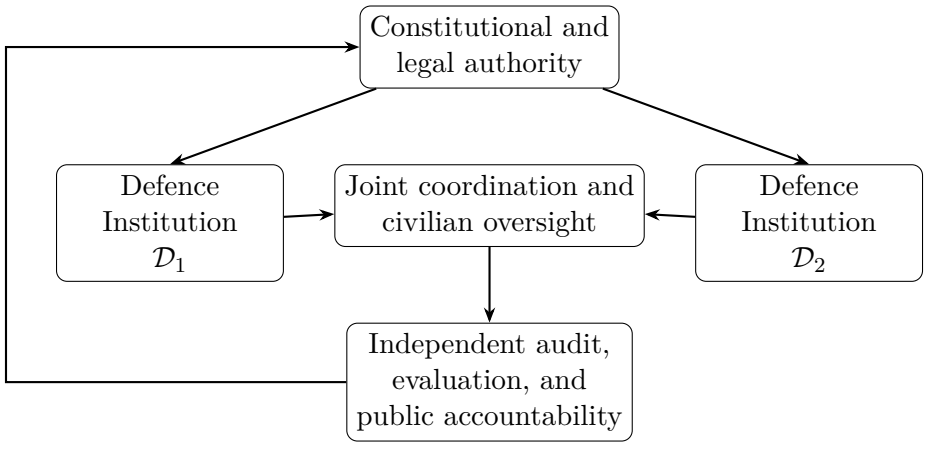


Figure 1: Conceptual governance architecture. The diagram expresses institutional accountability, not an existing organisational chart or a recommendation to create a particular force structure.

The architecture places legal authority above the two institutions and requires evaluation to feed back into governance. It does not assume that creating two institutions is inherently more efficient than creating one.

4 Economics and Financial Planning

4.1 Defence as a public expenditure problem

Defence is a public good in standard economic analysis, although specific services, facilities, and technologies may be congestible or excludable. The allocation of defence resources must therefore be considered alongside education, healthcare, infrastructure, social protection, and other public priorities (Samuelson, 1954).

Let B denote the available budget, and let c_i denote the planned cost of programme i. A basic budget constraint is

$$\sum_{i=1}^{N} c_i \leq B. \tag{21}$$

A responsible planning process should distinguish personnel, procurement, maintenance, training, pensions, cybersecurity, logistics, and long-term liabilities. Acquisition cost alone does not measure the full cost of a capability.

For a programme with initial investment I_0 and expected future net benefits $R_t - C_t$, a standard discounted valuation is

$$V = -I_0 + \sum_{t=1}^{T} \frac{\mathbb{E}[R_t - C_t]}{(1+r)^t}. \tag{22}$$

The discount rate r must reflect the valuation framework. For public projects, distributional effects, uncertainty, irreversibility, and intergenerational obligations may require analysis beyond a single net-present-value figure.

4.2 Institutional complementarity

Let V_1 and V_2 represent the assessed public value of two institutions and let C_{12} represent their coordination cost. A simplified joint-value expression is

$$V_{\text{joint}} = V_1 + V_2 + S_{12} - C_{12}, \tag{23}$$

where S_{12} is the incremental value attributable to complementarity, after accounting for overlap and interactions.

The dual structure is beneficial only if the joint arrangement outperforms feasible alternatives after accounting for its full costs, risks, and institutional consequences.

4.3 Financial risk and sensitivity analysis

A defence investment plan should be evaluated against uncertainty in inflation, interest rates, exchange rates, supplier concentration, delivery schedules, and maintenance costs.

For a financial quantity Y, sensitivity to an input parameter θ can be represented by

$$S_\theta = \frac{\partial Y}{\partial \theta}. \tag{24}$$

Scenario analysis should include baseline, adverse, and severe but plausible cases. Where uncertainty is substantial, decision makers should report ranges and assumptions rather than a single apparently precise forecast.

5 Econometrics, Statistics, and Data Science

5.1 A testable research design

The proposed framework should not be judged by the elegance of its polynomial representation. It should be assessed through measurable outcomes, defensible comparison groups, and reproducible statistical methods.

Possible outcome variables include:

- budget forecast errors;
- procurement delays and cost overruns;
- maintenance availability;
- audit findings and control failures;
- cybersecurity incident recovery times;
- civilian-harm prevention and legal compliance indicators;
- delivery of civilian public services under fiscal stress.

A panel-data specification might take the form

$$Y_{it} = \alpha_i + \lambda_t + \beta D_{it} + \boldsymbol{\gamma}^{\mathsf{T}} \mathbf{X}_{it} + \varepsilon_{it}, \tag{25}$$

where Y_{it} is an outcome for institution or jurisdiction i at time t, α_i captures unit-specific effects, λ_t captures common time effects, D_{it} measures a clearly defined institutional feature, and $\mathbf{X}_{it}$ contains justified controls.

The coefficient β is not automatically causal. Its causal interpretation depends on the research design and assumptions about confounding, selection, measurement, and reverse causality.

5.2 Uncertainty and validation

The empirical programme should include:

1. a preregistered hypothesis and outcome definition;
2. documented data provenance and missing-data treatment;
3. baseline models and relevant alternative specifications;
4. out-of-sample evaluation;
5. uncertainty intervals and sensitivity analysis;
6. independent replication where data access permits.

A mathematical identity cannot substitute for this process. The model's predictive value must be measured against simpler and more transparent alternatives.

6 Algorithms and Computational Implementation

A reproducible implementation can verify the polynomial identities and test flatness without relying on symbolic manipulation by hand.

6.1 Pseudocode

```
INPUT: Prime indices p = 2 and q = 113

FUNCTION cyclotomic_prime(p):
    RETURN sum(x^k for k = 0,...,p-1)

FUNCTION height(coefficients):
    RETURN max(abs(a) for a in coefficients)

P2   = cyclotomic_prime(2)
P113 = cyclotomic_prime(113)
Product = expand(P2 * P113)

ASSERT degree(P2) == 1
ASSERT degree(P113) == 112
ASSERT height(P2) == 1
ASSERT height(P113) == 1
ASSERT height(Product) == 2

OUTPUT polynomial identities and verification results
```

This pseudocode describes symbolic verification. A concrete implementation should specify the computer algebra system, coefficient representation, software version, and test environment.

6.2 Artificial intelligence and machine learning

Machine learning may assist with non-sensitive administrative tasks such as expenditure forecasting, maintenance planning, anomaly detection, and document classification. Such systems require validated data, documented performance, human review, and safeguards against automation bias.

A predictive model should be evaluated on held-out data using metrics appropriate to the task. For a continuous outcome, one possible metric is mean absolute error:

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N} |y_i - \widehat{y}_i|. \tag{26}$$

For high-consequence decisions, aggregate accuracy is insufficient. Calibration, subgroup performance, interpretability, data leakage, and the consequences of false positives and false negatives must also be assessed.

7 Computer Security and Cybersecurity

A dual institutional structure may introduce both resilience and additional attack surfaces. Separate administrative domains can reduce some common-mode risks, but duplicated identity systems, interfaces, suppliers, and poorly defined responsibilities may increase complexity.

A governance-level security programme should include:

- least-privilege access and multifactor authentication;
- secure software development and dependency management;
- independent logging and audit;

- incident response and recovery exercises;
- backup integrity and continuity planning;
- protection of classified, personal, and commercially sensitive information.

Cybersecurity should be evaluated through recognised risk-management practices rather than through a polynomial's coefficient structure. The NIST Cybersecurity Framework provides a reference for organising cybersecurity outcomes and risk management (NIST, 2024).

8 Physics and Complex Systems

Complex numbers are widely used in physics to represent oscillations, phase, and wave phenomena. For example, a sinusoidal signal can be represented by the real part of a complex exponential:

$$s(t) = \operatorname{Re}\left(Ae^{i\omega t}\right). \tag{27}$$

The mathematical usefulness of complex representation does not establish that a defence institution obeys the same physical equations. Such an application would require independently specified state variables, dynamics, units, and empirical tests.

For complex systems, useful questions include stability, sensitivity, feedback, delay, and the propagation of uncertainty. These concepts can inform institutional analysis without claiming that the cyclotomic polynomial is a physical law.

9 Chemistry, Biology, and Environmental Systems

Chemistry and biology contribute to defence preparedness through public health, environmental monitoring, materials safety, medical logistics, and the protection of civilian infrastructure.

A robust institutional framework should support:

- environmental impact assessment;
- safe handling and disposal of hazardous materials;
- disease surveillance and emergency medical capacity;
- protection of water, food, and energy systems;
- independent assessment of ecological consequences.

These applications require domain-specific scientific evidence. The polynomials in this treatise do not independently predict chemical reactions, biological outcomes, or environmental effects.

10 Linguistics, Psychology, and Psychiatry

Communication failures can undermine coordination even where formal responsibilities are well defined. Linguistics contributes to terminology management, translation quality, and the reduction of ambiguity in legal and technical documents.

Psychology informs the study of decision-making under uncertainty, group dynamics, cognitive bias, fatigue, and organisational learning. Psychiatry and occupational health contribute to the prevention and treatment of mental-health difficulties among personnel.

Appropriate institutional measures include clear reporting channels, confidential access to care, evidence-based training, and non-punitive mechanisms for reporting safety concerns. Individual diagnoses should not be inferred from institutional performance data or automated behavioural classifications.

11 Philosophy, Logic, and Religion

11.1 Logic and epistemic discipline

A valid mathematical deduction establishes what follows from specified premises. It does not establish that the premises adequately represent a political or military system.

The distinction can be expressed as

$$\text{Valid deduction from a model} \not\Rightarrow \text{Empirical validation of the model.} \tag{28}$$

This distinction is essential when formal mathematical structures are used to motivate institutional reforms.

11.2 Ethics and moral responsibility

Philosophical analysis should consider the relationship between means and ends, public accountability, proportionality, and the distribution of risks and benefits. Military effectiveness cannot be assessed solely by expenditure efficiency or technical capability.

Religious and philosophical traditions may provide resources for reflection on duty, restraint, compassion, and the value of human life. Their interpretation varies across traditions and communities; no single religious position should be presumed to represent all citizens.

12 Law, Constitutional Governance, and International Relations

12.1 Domestic legal authority

Any real institutional design must be consistent with the applicable constitution, legislation, civilian chain of command, appropriations process, and judicial review.

The following principles are central to accountable governance:

1. clearly defined statutory responsibilities;
2. civilian control and democratic accountability;
3. lawful procurement and expenditure controls;
4. independent audit and appropriate legislative scrutiny;
5. protection of fundamental rights and due process.

The framework presented here is conceptual and does not constitute a legal opinion on the constitutional status or feasibility of any particular proposed organisation.

12.2 International law

International legal obligations must be assessed independently of any mathematical or economic model. The Charter of the United Nations establishes the general framework governing the use of force and collective security. International humanitarian law governs the conduct of hostilities and the protection of persons in armed conflict.

Relevant primary sources include the United Nations Charter and the Geneva Conventions of 1949. Their applicability depends on the circumstances and the relevant legal rules.

No claimed strategic advantage, polynomial identity, or economic benefit overrides an applicable legal obligation.

13 Political Science, Geopolitics, and Diplomacy

The existence of complementary institutions does not by itself establish geopolitical superiority. Outcomes depend on political legitimacy, economic capacity, geography, alliances, technological development, public trust, and the behaviour of other states.

A serious comparative analysis should consider:

- regional security dilemmas and escalation risks;
- the costs and benefits of international cooperation;
- confidence-building and crisis communication;
- arms-control and non-proliferation obligations;
- the relationship between defence expenditure and sustainable economic development.

Diplomacy provides mechanisms for reducing uncertainty and resolving disputes without resorting to armed conflict. Strategic stability therefore requires more than institutional capability: it also requires communication, restraint, credible commitments, and mechanisms for managing disagreement (Schelling, 1966).

14 Welfare Economics and Warfare Economics

14.1 Social opportunity cost

Public expenditure on defence has an opportunity cost. The welfare implications of a budget depend on what is funded, what is displaced, who receives the benefits, and who bears the risks.

Let W denote a social-welfare function and let $\mathbf{g}$ be a vector of public expenditures. A simplified allocation problem is

$$\max_{\mathbf{g}} W(\mathbf{g}) \tag{29}$$

subject to

$$\sum_{i=1}^{N} g_i \leq B, \qquad g_i \geq 0. \tag{30}$$

The function W must explicitly represent the relevant social objectives and distributional judgments. There is no universally correct numerical weighting of security, health, education, liberty, environmental quality, and other public goods.

14.2 Warfare economics and prevention

Warfare economics studies the economic consequences of conflict, including destruction of productive capital, displacement, fiscal stress, trade disruption, and long-term reconstruction.

A broader public-policy assessment should compare prevention, preparedness, crisis management, and post-conflict recovery. The costs of conflict extend beyond direct military expenditure and may persist for generations.

A responsible evaluation therefore includes civilian protection, economic continuity, humanitarian assistance, and the opportunity costs of prolonged conflict.

15 Statistics, Risk, and Institutional Resilience

Risk analysis should distinguish probability, severity, uncertainty, and exposure. A simple expected-loss measure is

$$\mathbb{E}[L] = \sum_{j=1}^{J} p_j L_j, \tag{31}$$

where p_j denotes the probability of scenario j and L_j its associated loss.

Expected loss is not a complete measure of risk. Tail events, uncertain probabilities, correlated failures, and unequal distribution of harm may require additional measures.

Institutional resilience can be assessed through the ability to maintain essential public functions, recover from disruption, learn from failures, and preserve lawful accountability.

The analysis should avoid false precision. Where the underlying data are weak, qualitative uncertainty assessments may be more honest than elaborate numerical scores.

16 Research Programme and Falsifiable Hypotheses

The proposed framework generates questions rather than proven claims about defence institutions.

Table 2: Proposed research hypotheses

ID	Hypothesis	Evidence required
H1	Clear division of institutional responsibilities reduces administrative duplication.	Comparable expenditure and audit data.
H2	Joint oversight improves accountability.	Independent audit findings and governance measures.
H3	Integrated financial planning reduces forecast errors.	Out-of-sample budget forecast comparisons.
H4	Common cybersecurity standards improve recovery outcomes.	Documented incident and recovery data.
H5	The cyclotomic representation adds predictive value.	Out-of-sample performance against simpler baseline models.

Each hypothesis should be tested independently. A result supporting one hypothesis does not establish the others. In particular, the algebraic flatness of the two cyclotomic polynomials is not evidence that a dual institutional architecture is superior.

17 Limitations

The mathematical results in this treatise are exact. The proposed institutional interpretation is exploratory.

The polynomials $\Phi_2(x)$ and $\Phi_{113}(x)$ do not establish the optimal number of military organisations, determine force structure, predict conflict outcomes, or demonstrate an advantage for any country.

The index 113 has a mathematical role as a prime number. No independent political, geographical, demographic, or military significance follows from that fact.

Similarly, a complex-valued representation is useful only if its variables and operations have a defensible interpretation. A mathematical analogy should not be confused with an empirical mechanism.

Finally, the name "Greater India" is used as the title's proposed conceptual setting. It does not imply any territorial claim, political mandate, or endorsement of coercive expansion.

18 Conclusion

The two polynomials provide a compact and rigorous mathematical starting point:

$$\Phi_2(x) = x + 1, \tag{32}$$

$$\Phi_{113}(x) = 1 + x + x^2 + \cdots + x^{112}. \tag{33}$$

Both are flat and irreducible over the rational numbers. Their product, however, has height two and is not flat.

This result illustrates a general analytical caution: properties of individual components need not survive aggregation. In institutional design, coordination, overlap, and interaction must be evaluated explicitly rather than assumed away.

The proposed Complex Military Field Theory can be developed as a research programme connecting formal mathematics with public finance, governance, security, welfare, and empirical evaluation. Its practical value will depend on clear definitions, lawful institutions, reproducible evidence, and demonstrable improvements in public outcomes.

The ultimate objective of responsible defence planning should be the protection of people and lawful institutions, the preservation of peace where possible, and the sustainable use of public resources.

A Algebraic Verification

For a prime p, the factorisation of $x^p - 1$ gives

$$x^p - 1 = (x - 1)\Phi_p(x). \tag{34}$$

Dividing by $x - 1$ yields

$$\Phi_p(x) = \sum_{k=0}^{p-1} x^k. \tag{35}$$

For $p = 2$ and $p = 113$, every coefficient of the resulting polynomial is one. Therefore, both polynomials have height one.

Their product has coefficients obtained by convolution:

$$c_k = \sum_{j=0}^{k} a_j b_{k-j}, \tag{36}$$

where a_j and b_j are the coefficients of Φ_2 and Φ_{113}, respectively, with coefficients outside their degree ranges taken to be zero.

Consequently,

$$c_k = \begin{cases} 1, & k = 0, \\ 2, & 1 \le k \le 112, \\ 1, & k = 113. \end{cases} \tag{37}$$

Hence the product has height two, completing the verification.

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Glossary

Cyclotomic polynomial
The polynomial whose roots are the primitive roots of unity of a specified order.

Flat polynomial
A nonzero polynomial whose integer coefficients belong to $\{-1, 0, 1\}$.

Polynomial height
The maximum absolute value of a polynomial's coefficients.

Complex field
In this treatise, a proposed analytical representation using complex-valued variables. It is not automatically a physical field or a validated theory of defence.

Institutional complementarity
A situation in which two institutions create additional value through coordination, after accounting for duplication and coordination costs.

Defence economics
The study of defence expenditure, resource allocation, procurement, security provision, and their economic effects.

Welfare economics
The analysis of resource allocation and its implications for social welfare and distribution.

Warfare economics
The study of the economic causes, costs, consequences, and reconstruction effects of armed conflict.

Econometrics
The application of statistical methods to economic data and hypotheses.

Out-of-sample validation
Evaluation of a model using observations not used to estimate its parameters.

Institutional resilience
The capacity of an organisation or system to maintain essential functions, recover from disruption, and learn from failure.

Civilian oversight
Legal and institutional mechanisms through which civilian authorities supervise defence organisations and their expenditure.

The End